ISOLATED FLYBACK CONVERTER

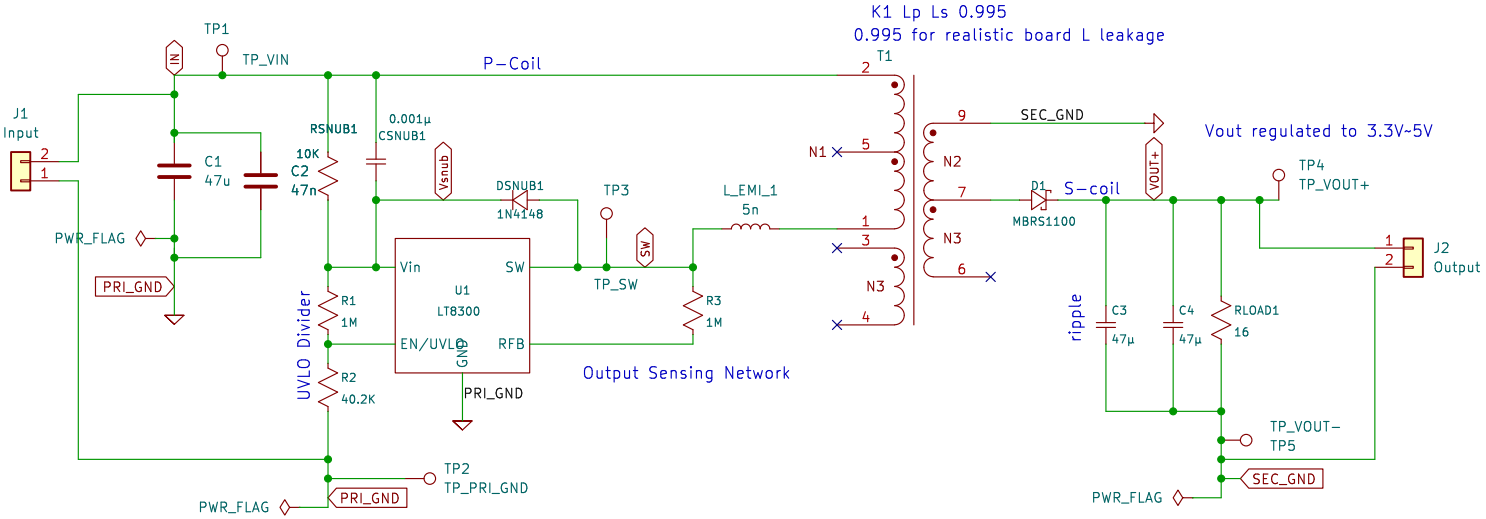
SPECS

36 < Vin < 50
Vout ≈ 5 V
Iout ≈ 0.4 A (DCM-ish Operation Mode)

NOTES

GND isolated from p-coil and s-coil
K1 is a statement for a fake transformer, acting like a coupled inductor
Cin filters incoming ripple
Cout reduces output ripple
R1-R2 divider based on datasheet suggestions
Rsnub, Csnub, Dsnub represent a RCD clamp
T ratio = 4
D1 is rev-biased when LT8300 MOSFET is ON, FW-biased when OFF
FB resistor R3 based on datasheet suggestions
Simulation Mode: Transient Simulation with 10ms with startup
Dsnub and D1 are rated to ~ 100Vbr
0.995 in the K-statement represents true tranformer leakage inductance
Additional EMI inductor added in series to p-coil for board SW node parasitics

NC = Not Connected
TPs added to probe oscilloscope signals
Transient Simulation Completed Fully in LTspice



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